Quantum Computing - Semester Guide Book

1. Course Objective

This course provides a strong mathematical foundation for quantum computing. It helps students understand how quantum systems work, interpret various applications, and analyze factors affecting computation in quantum environments.

2. Physical Properties and Mathematical Foundations

Quantum computing is based on principles of quantum mechanics.

Key topics include Double Slit Experiment, Particle vs Wave nature of light, Heisenberg Uncertainty Principle, Vector spaces, Inner and Outer products, Tensor products, and Linear operators.

These concepts form the backbone of quantum state representation and transformations.

3. Quantum Computing Postulates and Gates

Quantum systems follow specific postulates defining state, measurement, and evolution.

Important concepts include Bloch sphere visualization, single and multi-qubit systems, superposition, entanglement, Bell states, and quantum gates like Pauli, CNOT, and phase gates.

These gates manipulate qubits and enable quantum computations.

4. Quantum Computing Circuits

Quantum circuits are built using quantum gates.

Dirac notation is used to represent quantum states. Concepts include computational basis, orthonormality, Hadamard gates, and phase gates.

IBM Q Composer is used to simulate circuits and understand qubit transformations.

5. Fundamental Quantum Algorithms

Quantum algorithms provide speed advantages over classical algorithms.

Deutsch-Jozsa algorithm determines function type efficiently.

Grover’s algorithm is used for searching unsorted databases using amplitude amplification.

Concepts include oracle, diffuser, and multiple solution handling.

6. Programming on a Real Quantum Computer

Students learn to program real quantum computers using IBM Q.

They perform experiments, measure quantum states, and analyze outputs.

This bridges theoretical concepts with real-world quantum hardware.